

Topic 2 – Programming and development

Students should:		Curriculum reference
2.1	Be able to write and interpret programs in a high-level textual programming language. Any code in questions will use Python 3 code. See further information in Appendix 2	PS7.1A PS8.1A PS9.1A
2.2	Understand the purpose and use of variables, sequence, selection and repetition/iteration when writing programs	PD7.6B PD8.1B
2.3	Understand the need for and how to use data structures (records, one-dimensional arrays/lists)	PD8.2A
2.4	Understand the need for and how to manipulate strings	PD7.3A, B, C, D, E, F, G
2.5	Understand the purpose of subprograms and the role of arguments, parameters, return values and local variables in subprograms	PD7.7A PD8.2B PD9.2A, B
2.6	Understand the purpose of, and how to use, arithmetic operators (add, subtract, divide, multiply, modulus, integer division) and the order of operations (BIDMAS)	PD7.4A, B
2.7	Understand the purpose of, and how to use, relational operators (equal to, less than, greater than, not equal to, less than or equal to, greater than or equal to)	PD7.5A
2.8	Understand the purpose of, and how to use, logical operators (AND, OR, NOT)	PD8.3A
2.9	Understand the need for and use of pre-existing (built-in, library) subprograms in high-level programming languages	PD7.7A
2.10	Be able to locate and fix logical and syntax errors in a program	PD7.2A, B PD8.4A PD9.3A